

Research Article

Enhancing the Achievement of Learning Outcomes for Foreign Language Learners Based on Gamification and Card Games

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Gamification and card games have gained increased popularity across diverse educational domains, particularly in the field of foreign language learning (FLL). They offer potent tools that can transform the learning process into an engaging, interactive, and enjoyable experience. Currently, numerous instances of gamified or card game-based language learning methods have demonstrated their significant impact on sustaining learners' engagement and advancing their proficiency levels. However, it is noteworthy that these approaches often prioritize the motivational aspect while neglecting the imperative need to enhance student learning outcomes (SLOs). The latter holds pivotal importance not only in determining the success of academic courses but also in evaluating the effectiveness of e-content. To address this gap, the primary objective of this work is to enhance the achievement of SLOs within the context of foreign language courses. A novel framework is proposed for learning foreign languages, harnessing the potential of gamification and card games. This framework, developed through the integration of contemporary technologies used in e-learning systems and foreign language applications, encompasses agent technology and the microlearning strategy. A case study was conducted to evaluate the effectiveness of the proposed framework. The results revealed a statistically significant improvement in SLOs. This study contributes significantly to the advancement of both gamification and game-based learning, presenting a promising pedagogical paradigm for language teaching and learning.

Keywords: card games; gamification; language learning; learning outcomes; student achievement

1. Introduction

Owing to technology's capacity to transcend geographical boundaries and facilitate universal access to information and interpersonal connections, there exists an unprecedented demand for acquiring proficiency in foreign languages [1]. Over time, this demand has transformed into a substantial and fundamental skill for individuals. The acquisition of foreign languages offers a myriad of advantages, including the enhancement of skills, expansion of cognitive capacities, facilitation of global communication, creation of new career opportunities, improvement of self-esteem, and other noteworthy benefits [2]. In the contemporary era, digital education has become an indispensable component of the educational landscape. Modern learners, deeply immersed in technology, regard it as an essential tool for learning [3]. Evidently, advanced technology exerts a sub-

stantial positive influence on language learning. One such advancement is the utilization of gamification [3] and games [4]. Gamification involves integrating game mechanics into nongaming contexts. It is primarily focused on motivating learners and engaging them by incorporating game elements such as points, leaderboards, immediate feedback, badges, and progressing through game levels [5].

Notably, there is a growing body of research on the implementation of gamification in language learning, gaining popularity for motivating students, enhancing performance, and fostering enjoyment in computer-mediated and nongaming settings [1]. Furthermore, scholarly research underscores the efficacy of games as impactful learning instruments across diverse educational levels [6]. This effectiveness stems from the creation of a "play and learn" environment that heightens learners' interest in mastering the target language [6]. Card games are an example of successful

games employed in foreign language learning (FLL). These games provide a tactile and interactive learning experience, thereby augmenting the enjoyment and memorability of the language learning process [4]. Moreover, the incorporation of card games in language learning signifies an inventive and engaging approach capable of enriching various facets of language acquisition [7].

However, the majority of language learning approaches that have been proposed based on gamification or card games emphasize motivational aspects, often overlooking the essential enhancement of student learning outcomes (SLOs) [8]. Interestingly, SLOs are integral components of the educational framework, providing a structured approach to measuring, enhancing, and ensuring the quality of language education. They not only guide the learning process but also contribute to the overall success and satisfaction of students in their learning endeavors [9]. The sense of accomplishment derived from meeting specific learning outcomes serves as a powerful motivator, signifying student progress and growth over the course of their education. It indicates that students are advancing in their understanding and application of the subject matter, including language skills. Furthermore, SLOs ensure the quality of education and provide evidence that instructional methods, curriculum, and assessment strategies are effective in facilitating meaningful learning experiences [10]. Accordingly, there is a call for the development of approaches that holistically prioritize and optimize the achievement of SLOs. Hence, the primary objective of this work is to enhance the achievement of students' learning outcomes in foreign language courses. In pursuit of this objective, this paper introduces a novel framework for learning foreign languages based on gamification and card games. The development of the proposed framework leverages the potential of artificial intelligence (AI), including agent technology [10]. Additionally, it embraces the widely recognized microlearning approach [11], which has garnered considerable prominence in the realm of language education.

A case study was conducted to evaluate the effectiveness of the proposed framework in enhancing SLOs. The results revealed a statistically significant improvement in SLOs, underscoring the potential efficacy of gamification and card game-based approaches. This endeavor makes a substantial contribution to the progression of both gamification and game-based learning (GBL), introducing a promising pedagogical paradigm for the domain of language teaching and learning.

2. Related Work

This section encompasses various salient themes, namely, SLOs, agent technology, microlearning, gamification in the context of FLL, and card games in the context of FLL.

2.1. SLOs. In the current educational landscape, there is a noticeable shift toward emphasizing SLOs. The SLOs represent articulated statements outlining the anticipated achievements of a successful student or learner upon completing a specific program module, course unit, or qualification [10].

The core objectives of SLOs can be defined in key dimensions [12]:

1. *Informed student progression:* SLOs serve to inform students about the knowledge and skills they are expected to gain through a specific course or program of study.
2. *Holistic mapping of educational trajectories:* SLOs contribute to mapping the intricate relationships between various courses, programs of study, and academic degrees, offering a comprehensive view of educational trajectories.
3. *Curriculum development and progression:* SLOs play a crucial role in mapping the development of knowledge and skills at each curriculum level, ensuring a coherent and progressive learning experience.
4. *Structural framework for assessment:* SLOs offer a structured framework for evaluating teaching and learning processes, providing a foundation for meaningful assessment strategies.
5. *Guidance for curriculum design and teaching practices:* SLOs inform curriculum design and pedagogic practices, guiding educators in creating effective and relevant educational experiences.
6. *Communication of performance standards:* SLOs serve as a means of communicating clear and specific standards of performance expected from students, setting a benchmark for achievement.

In the realm of language learning, the realization of SLOs holds paramount significance, serving as an indication of the triumphant acquisition and mastery of indispensable language skills and competencies. These outcomes, meticulously delineated by educational institutions, encompass a spectrum of linguistic abilities, including proficiency in speaking (s), listening (l), reading (r), and writing (w). The accomplishment of these learning objectives ensures that students not only comprehend the foundational principles of the language but also cultivate the practical skills essential for proficient communication [13].

Moreover, attaining these objectives demonstrates a student's preparedness to navigate real-world situations in the target language, augmenting their comprehensive language proficiency and cultural acumen. Ultimately, the emphasis on the fulfillment of SLOs contributes substantively to the efficacy and triumph of foreign language education [14]. In light of this, the primary objective of this paper is to improve the achievement of SLOs among individuals engaged in FLL.

2.2. Agent Technology. Agent technology, stemming from the fields of AI and computer science, has emerged as a transformative force in various domains. An agent is a software entity endowed with autonomy, capable of performing activities on behalf of a user or another program while holding knowledge or representations of the user's goals or intentions [15]. The functioning of an agent-based system relies

on interactions between autonomous, loosely connected software or hardware units known as agents. These agents are characterized by their mobility, the ability to communicate independently of the user's presence, and adaptive thinking [16].

Agents exhibit key attributes, including autonomy, reactivity, proactiveness, social ability, cooperative capacity, reasoning skills, and adaptability. These attributes make agent technology a subject of investigation in various scientific disciplines, highlighting its potential to revolutionize the educational landscape [15]. In the realm of e-learning, agent technology plays a crucial role in reshaping and enhancing the learning experience for students and educators alike. E-learning systems face numerous challenges, such as managing extensive learning content, diverse formats, multiple access methods, and adherence to standards. Additionally, selecting material aligned with learners' needs, including prior knowledge, preferences, learning styles, personalities, and interests, presents a significant challenge. The application of agents in e-learning environments can address these challenges by supporting intelligent learning. Agents can assist instructors and guide learners to achieve educational objectives by considering individual accomplishments, advancements, and preferred learning approaches. This personalized and adaptive approach enhances the overall learning experience by providing tailored support for both educators and learners [17]. Accordingly, the agent serves as the foundational component of the language learning framework proposed in this work.

2.3. Microlearning. The concept of microlearning has garnered considerable attention in the realm of language learning, contributing to an engaging and fulfilling educational experience. It presents information in bite-sized, easily digestible units, promoting effective retention and recall. Microlearning demonstrates its effectiveness in fostering the development of language skills, encompassing areas such as vocabulary and grammar, concurrently facilitating the acquisition and retention of instructional content [18].

Beyond improved memory retention, this approach exhibits adaptability to individuals with diverse proficiency levels, as it does not necessitate prolonged periods of concentrated focus. Moreover, it provides a satisfactory level of accessibility, thereby mitigating the perceived challenges of learning a new language. The incorporation of microlearning often involves the dissemination of information through online platforms, empowering learners to partake in self-directed study [19]. Such an approach not only augments learning outcomes but also contributes to time and cost efficiency by diminishing reliance on conventional teaching methods and physical classrooms [20]. Considering these advantages, the framework presented in this study aligns with the principles of the microlearning approach for the delivery of educational content.

2.4. Gamification in the Context of FLL. In the ever-evolving landscape of education, traditional methods are constantly being reimagined to meet the needs of modern learners. One such transformative approach gaining momentum is

gamification [3]. Gamification refers to the incorporation of game elements into nongame contexts [21]. In the domain of FLL, the utilization of gamification has become a subject of increasing interest and exploration in recent years, as it brings about numerous benefits that make the learning process more engaging and enjoyable. One fundamental benefit is the enhancement of motivation. By introducing elements of challenge, competition, and reward, gamification taps into learners' motivation, making language acquisition a more interactive and purposeful endeavor [22]. Additionally, gamification fosters a sense of achievement and progress. As learners accomplish tasks, unlock levels, or earn rewards, they experience a tangible sense of accomplishment. This not only boosts their confidence but also encourages a positive attitude toward language learning. The element of enjoyment is another key benefit. Gamified language learning platforms often create a more dynamic and entertaining environment, transforming potentially mundane language exercises into engaging activities. This shift in tone contributes to a more positive and enthusiastic approach to learning. Gamification also promotes a sense of community. Whether through collaborative challenges, group activities, or shared achievements, learners can connect with each other, fostering a supportive learning environment. Furthermore, gamification facilitates personalized learning experiences. Learners can choose activities, challenges, or themes that align with their interests and preferences, allowing for a more tailored approach to language acquisition. This personalization increases engagement and makes the learning journey more relevant to individual learners [3].

Several studies have embraced gamification as a valuable approach to enriching the language learning experience. For example, the authors in [23] designed a digital resource tool to reinforce theoretical knowledge acquired within a virtual language learning environment. The tool included various tasks, gamification elements, and competitive features. According to the assessment, it exhibited substantial promise as a valuable asset for enhancing the language learning environment. The authors in [24] conducted an empirical study to examine the influence of gamification on the time-limited performance of English majors in Chinese colleges. They utilized a gamified platform with diverse game components like rankings, points, levels, and leaderboards. Results indicated that integrating game elements had the potential to captivate learners and enhance their proficiency in English. The authors in [25] conducted a thorough analysis of three websites aimed at incorporating new elements within a conventional classroom environment. The results of this analysis revealed that, despite the distinct features inherent in these tools, all three demonstrated the capacity to elevate levels of engagement and motivation.

Additionally, gamification is widely implemented in mobile applications, aiding vocabulary memorization and language skill acquisition. These apps not only provide an engaging user experience but also enhance financial gains for developers [26]. The market has witnessed the introduction of numerous gamified applications, with Kahoot! emerging as a prominent free gamified mobile platform for

language learning [27]. Kahoot! is a popular gamified mobile platform, incorporating elements like points and leaderboards. It transforms classrooms into game-show environments, contributing to an entertaining and motivational learning experience. Numerous scholarly studies have underscored the benefits of incorporating Kahoot! in language school classrooms, emphasizing its popularity as a student gamification application featuring various engaging elements for both educators and learners [28]. On the other hand, numerous researchers have undertaken a systematic literature review to examine the impact of gamification on the pedagogy of acquiring new languages [29]. The results highlighted the effectiveness of gamification as a valuable tool in language education, showcasing its ability to boost learners' motivation and reshape the learning experience into an enjoyable and engaging process.

Remarkably, a predominant focus of gamification initiatives in language learning has been on elevating student motivation, often overlooking the crucial aspect of realizing SLOs. Although motivating students through gamification is valuable, it is imperative to align these motivational strategies with the achievement of predefined SLOs. As illustrated in Section 4.1, the realization of SLOs ensures that language learners not only sustain engagement and motivation but also acquire the essential linguistic and cultural competencies for meaningful communication and success in academic and professional pursuits. In addressing this notable gap, the present paper introduces a gamified language learning framework explicitly designed to optimize the attainment of SLOs.

2.5. Card Games in the Context of FLL. The field of FLL has undergone a notable shift in pedagogical approaches, with growing recognition among educators of the value of incorporating innovative and engaging methods [29]. Among these methods, the integration of card games into language learning environments has gained attention for its potential to enhance linguistic proficiency, cultural understanding, and overall learning experiences [30]. Card games, beyond their recreational value, serve as effective learning tools by engaging learners in meaningful communication and fostering language production and comprehension skills. Their interactive nature not only enhances motivation and engagement but also stimulates strategic thinking, problem solving, and memory recall, creating a conducive environment for language learning [31].

The evolution in foreign language education has prompted researchers and educators to investigate the efficacy of integrating card games into language learning curricula. For example, but not limited to, the authors in [7] explored the U-NO-ME card game's pedagogical utility in improving English vocabulary acquisition among primary school students. This study enhanced U-NO-ME cards through visuals and power cards, demonstrating positive impacts on vocabulary retention and motivation. The authors in [4] introduced the Vocabulary Connect 4 Circles (VC4C) educational tool, drawing inspiration from two-player connection games. VC4C positively influenced students' vocabulary skills, fostering improvements in coopera-

tion, communication, creativity, and critical thinking. The authors in [30] presented a pedagogical application rooted in card game dynamics, aiming to enhance children's English vocabulary proficiency. This app, tailored to distinct proficiency levels, evidenced marked improvements in children's vocabulary skills. Moreover, the authors in [32] offered inclusive insights into how learners' use of word cards contributes to vocabulary learning. Overall, their study favored paperwork cards over digital ones and ready-made cards over self-constructed ones, with a tendency for more substantial effects in intentional and massed learning conditions.

In the realm of grammar skills acquisition, the authors in [31] conducted a study involving a language card game inspired by Donkey and Joker games. This game, consisting of 53 cards, was aimed at assisting students in learning and applying adjectives while reducing anxiety in English learning. This game demonstrated efficacy in enhancing students' interest in grammar intricacies. Furthermore, the authors in [33] investigated the impact of a card game intervention on improving English verbal communication skills for nursing students from diverse backgrounds. Findings revealed three key themes: education content, skills development, and fun and creativity, emphasizing the importance of creative practice in a safe learning environment.

While previous research has predominantly focused on the motivational aspects of card game integration, elucidating how these games can engage and stimulate students in the learning process, this focus on motivation is undeniably valuable. However, there appears to be a discernible gap in the literature concerning the strategies and methodologies employed to enhance the achievement of SLOs. Addressing this gap, the study introduces a framework utilizing card games to enhance the achievement of SLOs in language learning.

3. Proposed Framework

For the time being, the educational process is moving in the direction of a learning outcome-based approach [10]. Hence, all learning activities should be closely aligned with learning outcomes. This work is aimed at enhancing the student's achievement of learning outcomes related to FLL. To meet this aim, we combine the idea of educational CGs with point-game elements and propose a framework for learning foreign languages. As illustrated in Section 2.5, card games play a prominent role in motivating learning. In addition, several studies have shown that points are one of the most commonly used game elements in the learning domain [34]. Figure 1 shows the proposed framework. It consists of the card drafting agent (CDA) and the card set database. The description of the CDA and the card set database is provided in Sections 3.1 and 3.2, respectively. Interestingly, the framework can be used as a standalone language learning tool or integrated to support existing language learning platforms.

3.1. CDA. Agent technology proves particularly efficacious in tackling applications fraught with significant challenges,

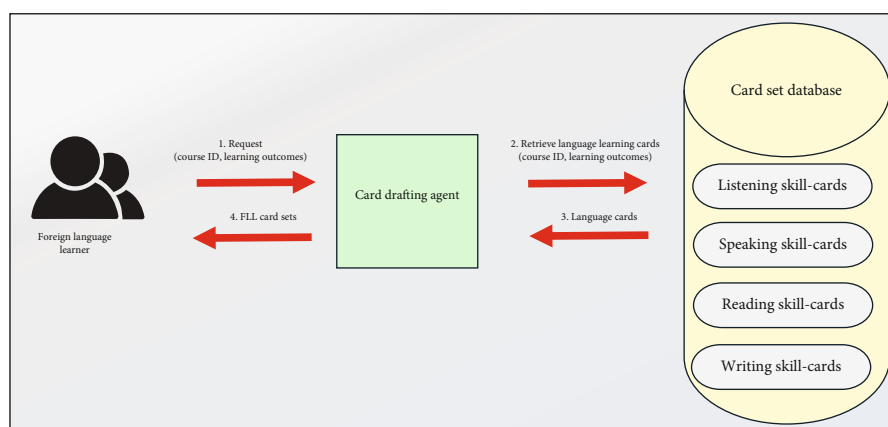


FIGURE 1: The proposed framework.

such as geographical distance, collaboration across diverse entities, and the integration of multiple software components [35]. Consequently, a pivotal component within the proposed framework is the CDA. The CDA assumes responsibility for crafting card sets pertinent to various learning outcomes in a specific language course. The structural depiction of this agent is delineated in Figure 2, while its behavior is elucidated in Figure 3. As depicted in Figure 3, the CDA responds to a student's request, comprising a course ID and the SLOs to be achieved. Subsequently, it retrieves pertinent cards from the card set database and formulates the FLL card set corresponding to each SLO. Ultimately, the CDA forwards the generated FLL card set to the student. The intricate details of the card structure are expounded upon in Section 3.3, with the card set generation process explicated in Section 3.4.

3.2. Card Set Database. Figure 4 illustrates the entity relationship diagram (ERD) of the card set database. It encompasses the learning cards essential for achieving diverse SLOs in the designated language course. These cards are categorized into five distinct classes: l skill-related cards (C_l), s skill-related cards (C_s), r skill-related cards (C_r), w skill-related cards (C_w), and prize cards (C_p). The CDA will access these cards to generate the FLL card sets tailored to various SLOs.

3.3. The Structure of the Card Set. Significant and efficacious education in foreign language courses focuses on four fundamental language skills: l, s, r, and w. These skills constitute an essential component of language education [36]. Consequently, each card set cs_j comprises five cards, accounting for the basic skills in FLL, along with an additional card designated as a prize card.

$$|C_x| \in CS_j = 5, \quad \text{where } 1 \leq x \leq 5 \quad (1)$$

The incorporation of a prize card within the card set is aimed at augmenting engagement levels during the learning process. Scholarly research indicates that students respond positively to external motivators, such as awards, leading to the development of internalized and robust incentives,

thereby prompting increased effort [37]. In this context, the prize card has the potential to serve as a viable form of reward. When students opt for the prize card, they encounter valuable educational information conducive to enhancing their achievement of the learning outcomes. This has the potential to cultivate enthusiasm and motivate students to progress further. The structural representation of a card set is depicted in Figure 5, with the intention of elevating the learner's achievement across various learning outcomes.

Each FLL card set comprises five cards, with four containing language learning content and a corresponding quiz for the student to answer. The fifth card serves as the prize card, exclusively offering language learning content tied to a specific learning outcome. The organization of content on each card aligns with the microlearning technique elucidated in Section 2.3. Cards related to l skills incorporate an audio recording pertaining to a specific learning outcome and a corresponding question for the student to solve. The student is required to listen to the recording and select the answer from the multiple choices presented on the screen. Cards related to s, r, and w skills consist of learning statements followed by relevant multiple-choice questions for students to address. Figure 6 illustrates the structure of each card in the FLL card set.

3.4. Card Set Generation Process. As previously elucidated, the CDA assumes responsibility for generating card sets associated with various SLOs. Upon receiving a student's request, the CDA retrieves the required cards from the card set database and subsequently assembles a card set wherein the cards are arranged in a randomized sequence. Algorithm 1 delineates the sequential procedures involved in generating a card set corresponding to each SLO.

3.5. Card Selection Rules. Figure 7 delineates the activity diagram of the card selection process. The student is obligated to randomly choose a card from the FLL card set, grasp its valuable information, and provide a response to the associated question. In the event of an incorrect answer, the student should select another card from the same card set. Conversely, a correct answer will grant the student a point as a reward and move him into the subsequent card set

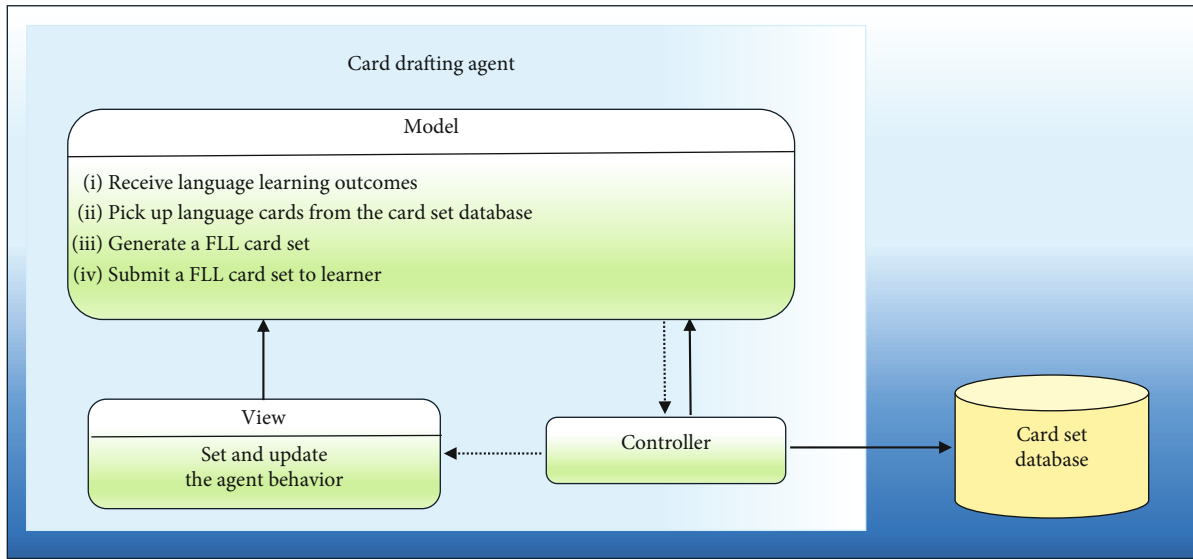


FIGURE 2: The structure of the card drafting agent.

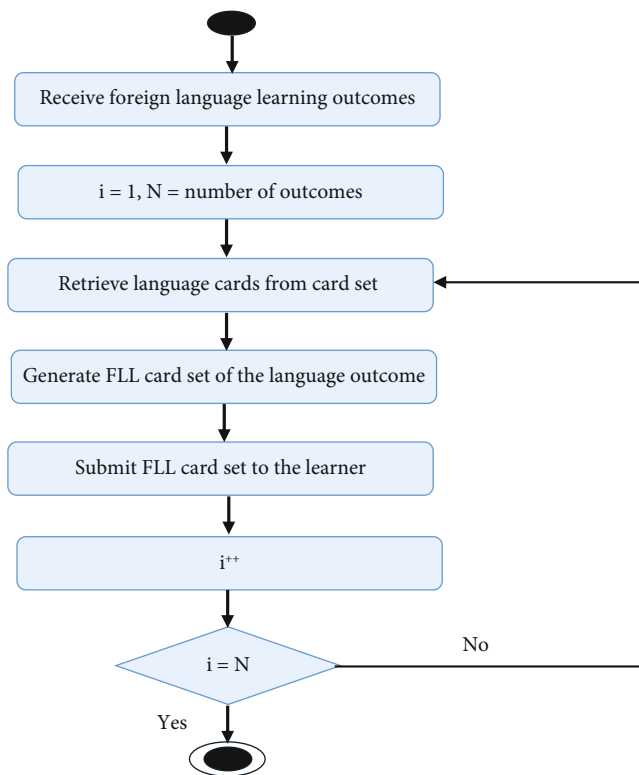


FIGURE 3: The description of the card drafting agent's behavior.

linked with another SLO. If the chosen card happens to be a prize card, the student is anticipated to comprehend its content and proceed to the succeeding card set.

4. Implementation and Evaluation of the Proposed Framework

4.1. Implementation. The proposed framework was implemented to fulfill the evaluation purpose. The Java Agent

Development Framework (JADE) [38] was utilized for the implementation of the CDA. JADE is a robust software framework designed for developing intelligent agents using the Java programming language. It provides essential tools for constructing and programming agent systems. The choice of JADE was motivated by its simplification of the primitives necessary for programming the multiagent system within the Java programming language, offering a resilient and robust environment conducive to applications based on object-oriented design. Additionally, JADE adheres to the specifications set forth by the Foundation for Intelligent Physical Agents (FIPA), facilitating communication between agents and enabling service detection within the system. FIPA has formulated specifications for interaction protocols, communicative acts, and content messages for agent communication, thereby supporting the potential extensibility of our proposed framework to incorporate additional functionalities in the future. Moreover, the web pages for our framework were developed using Java Server Pages (JSP) technology [39], and the card set database was implemented using MySQL Server [40].

4.2. Evaluation of the Proposed Framework

4.2.1. Research Methodology. This section delineates the study conducted to assess the efficacy of the proposed framework. The study adopted a quasiexperimental design [41], which facilitated the analysis of the influence of an independent variable (the FLL framework) on dependent variables (achievement of SLOs). Within this methodological framework, one of the experimental approaches was employed, entailing the establishment of both a control group and an experimental group. The sample size comprised 120 selected students, who were subsequently divided into two distinct groups. The experimental group was exposed to the independent variable (FLL framework), while this variable was intentionally withheld from the control group.

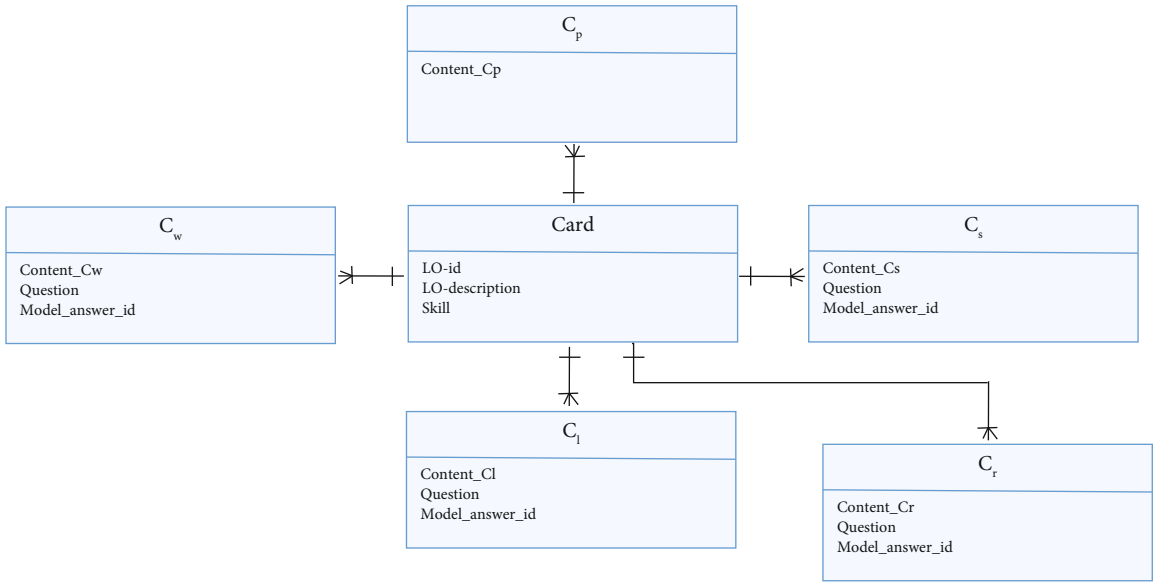


FIGURE 4: ERD of the card set database.

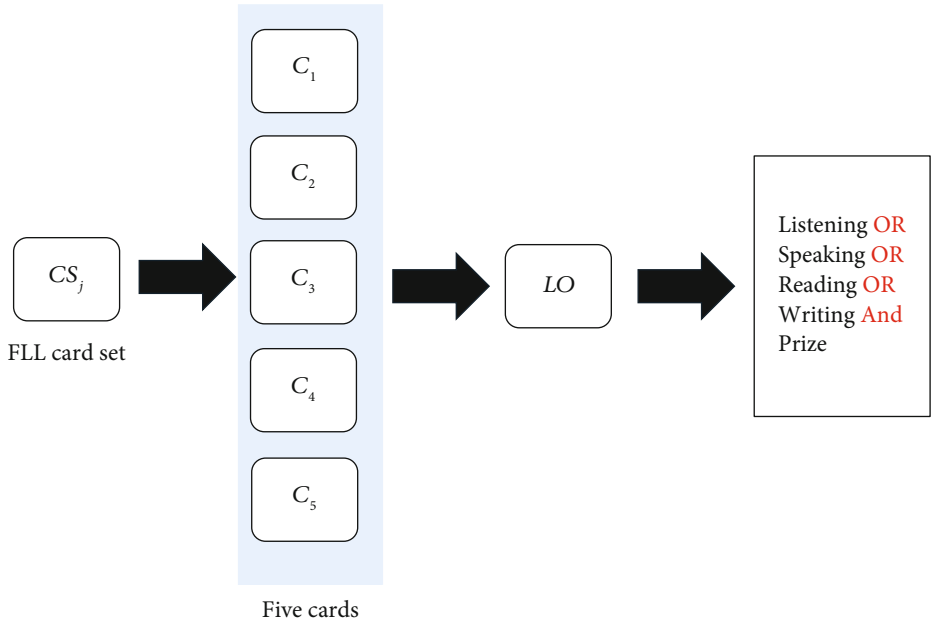


FIGURE 5: The structure of a card set.

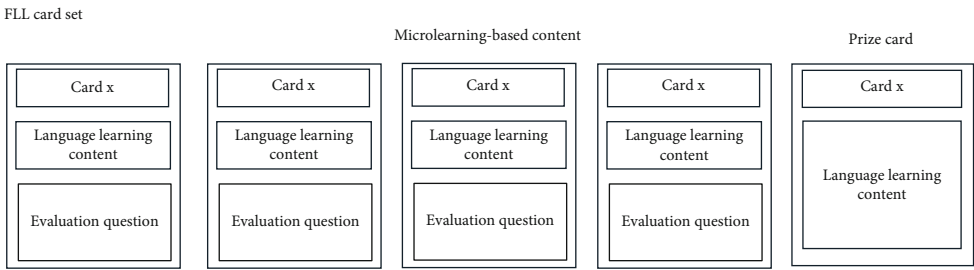
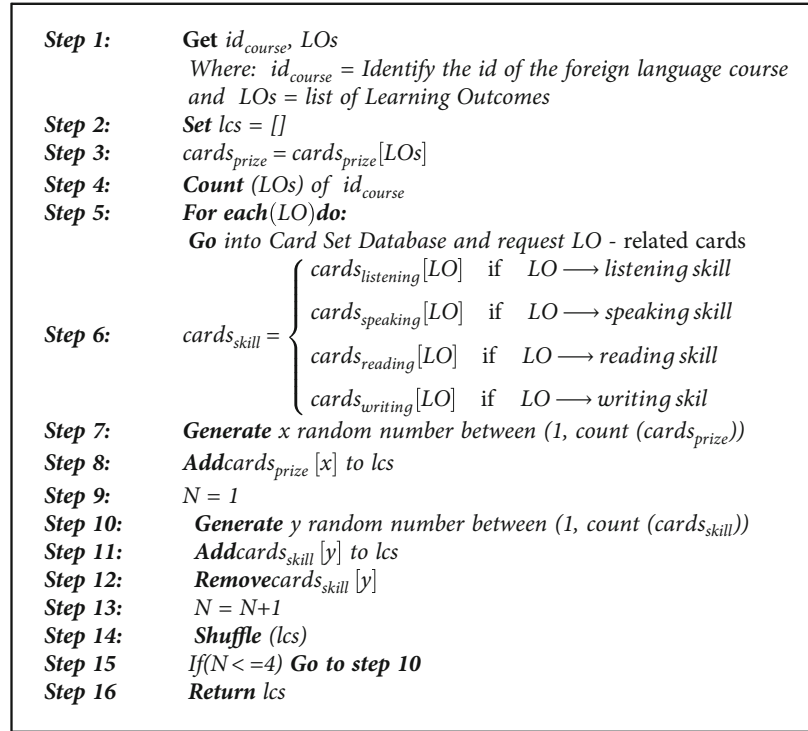


FIGURE 6: The structure of cards in the FLL card set.



ALGORITHM 1: Process for generating the card set.

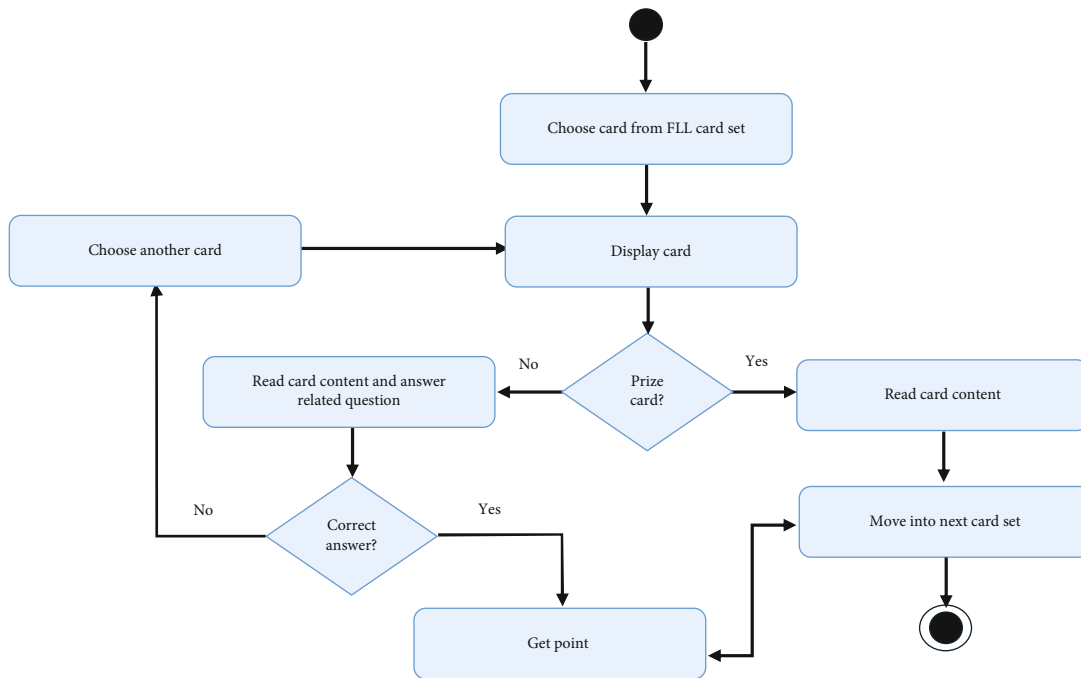


FIGURE 7: UML activity diagram of the card selection process.

Subsequently, postmeasurements were administered to both the experimental and control groups, aimed at ascertaining the discernible impact of the implementation of the FLL framework on the student's achievement of learning outcomes.

4.2.2. Study Population and Sample. The present investigation enrolled participants from an intermediate school situated in Jouf, Saudi Arabia, focusing on students in the middle stage of their academic progression. Inclusion within the research cohort was predicated upon satisfying

two specific criteria. Firstly, participants were deliberately selected from the demographic aged between 12 and 15 years. This deliberate stratification facilitated a comprehensive evaluation of the framework's impact on learners spanning varying age brackets. Secondly, explicit parental authorization for their child's involvement in the research and their endorsement of participation in the language acquisition framework were solicited and obtained.

These criteria were met by a total of 120 students. The allocation of these students into the experimental group was determined through the expertise of a student advisor at the school. The advisor meticulously identified students who possessed the requisite computer and internet skills, which were deemed essential for the successful completion of the anticipated tasks. Additionally, efforts were made to include students from diverse dialect communities within the participant pool. This deliberate inclusion was aimed at capturing a nuanced understanding of the proposed framework's efficacy across various linguistic backgrounds and significantly contributed to enhancing the ecological validity of the study. Furthermore, the control group, which did not receive the FLL system, was also selected based on the recommendations of the student advisor. Consequently, the sample was divided into an experimental group and a control group, each comprising 60 students.

4.2.3. Instruments. The instruments utilized in this study encompass subject-completed assessments, namely, achievement tests and surveys. Achievement tests serve the purpose of evaluating an individual's level of knowledge or proficiency within a specific subject area. They are traditionally administered in educational institutions to measure learning or the effectiveness of instructional practices [42]. Our study employed quizzes and an evaluation test as achievement tests. Quizzes gauge students' understanding during the learning process. For the experimental group, quizzes were integrated into the proposed system. Each card, excluding the prize card, provides learning material and a pertinent quiz involving a single multiple-choice question. Meanwhile, the control group underwent a parallel assessment process. Similar single multiple-choice questions were administered to students in the control group, but these assessments were delivered through the traditional method of paper-based quizzes. In addition, the evaluation of students' overall achievement of learning outcomes was conducted at the culmination of each academic period. This evaluation consisted of a standardized test comprising 20 multiple-choice questions, which were thoughtfully crafted by the course instructor. Students were required to respond to these questions within a specified 30-min timeframe. Importantly, the mode of delivery for these achievement tests diverged for the two groups under investigation. Specifically, the experimental group underwent online testing, facilitated by the integration of the e-learning system, while the control group engaged in paper-based testing.

Another pivotal instrument involves the utilization of surveys. Surveys provide a structured and systematic methodology for gathering data from a broad and diverse range

of participants, thereby facilitating the simultaneous acquisition of quantitative and qualitative information. Thus, upon the culmination of the study, participants belonging to the experimental group were invited to complete an online survey employing the conventional 5-level Likert scale, which spans from *strongly agree* to *strongly disagree*. This survey encompasses two inquiries, each capturing students' perspectives on the proposed framework from distinct vantage points: motivation and dialect variation. Recognizing the significant role of motivation in FLL success [43], participants were queried regarding whether the framework positively influenced their motivation to enhance their linguistic proficiencies. Additionally, acknowledging the multifaceted nature of dialects within communities and the ongoing debate surrounding dialect inclusion in language learning systems [44], the second inquiry delves into the potential impact of integrating students' native dialects within the proposed framework. Meanwhile, participants from the control group were presented with identical questions, albeit in a paper-based format, to ensure consistency in data collection procedures.

Moreover, at the study's conclusion, an online survey was administered to six English subject instructors who served as coordinators for the experiment. The survey featured a single question aimed at evaluating their perspectives on the applicability and reliability of the proposed solution. This feedback from instructors complements the student-centric evaluations, providing a comprehensive view of the framework's efficacy.

4.2.4. Data Collection Settings. The experiment took place during the second trimester of 2023, specifically divided into two periods: t_1 and t_2 , each lasting 6 weeks and comprising five 50-min learning sessions. The first period (t_1) commenced at the beginning of the trimester on November 26th, while the second period (t_2) started in the middle of the trimester on January 28th. The framework was tailored to deliver learning material and quizzes aligned with the English textbook taught during this trimester, utilizing the microlearning strategy. Throughout each period, close attention was given to monitoring the students' achievement of two designated learning outcomes:

- *Knowledge learning outcome (LO_1):* definition of basic features and terminology relevant to language skills.
- *Skill learning outcome (LO_2):* development of proficiency in the four language skills for effective communication.

The attainment of these outcomes played an integral role within the context of the English curriculum delivered during the trimester. During each period, participants underwent a series of quizzes and an evaluation test tailored to the specific content covered during that period. Additionally, during the 13th week of the study, participants in both groups engaged in the completion of a survey, encompassing dimensions such as motivation levels and considerations related to dialect variations. In parallel, English subject instructors contributed to the research by submitting an

additional online survey, reflecting their professional perspectives on the feasibility and applicability of the proposed solution.

4.2.5. Data Analysis. The progress of students in achieving learning outcomes LO₁ and LO₂ was assessed for both groups using the Continuous e-Assessment Method (CAM) algorithm, introduced in [10]. Primarily, CAM tracks and calculates students' achievement of learning outcomes, identifying areas of weakness, and suggesting corrective measures to enhance attainment. In this study, CAM was exclusively employed to calculate students' achievement of LO₁ and LO₂ during *t1* and *t2*. The utilization of CAM aligns seamlessly with the primary objective of this study, which is the enhancement of the achievement of SLOs. Additionally, the quantitative data obtained from surveys underwent thorough analysis using Microsoft Excel [45]. The widespread availability and basic statistical analysis capabilities of Excel contribute to its prevalent use for diverse data analysis tasks. In addition, Excel's charting and graphing tools were employed to create visual representations, aiding in the interpretation of trends and patterns. This analytical process involved meticulous examination and tabulation of participant responses.

4.2.6. Results. As a result of this experiment, our attention has been directed toward the following aspects:

1. The evaluation of students' enhancement in achieving learning outcomes (direct assessment) in the experimental group and control group
2. The perceptions of participants regarding the proposed framework, with emphasis on motivation and dialectal variation
3. The opinions of instructors about the applicability and reliability of the framework

In the context of learning outcome achievement, the CAM algorithm was employed to compute individual students' achievement throughout the experiment. Instructors provided comprehensive reports delineating students' achievements in LO₁ and LO₂. Presented in Table 1 are the cumulative achievements of LO₁ for each student in both the control and experimental groups during time periods *t1* and *t2*. Notably, 72% of the participants in the experimental group exhibited an enhancement in their achievement of LO₁ (positive increased rate [IR]), while 55% of the participants in the control group improved their achievement of LO₁. Table 2 displays students' achievements in LO₂ during *t1* and *t2*, revealing that 90% of the participants in the experimental group improved their attainment of LO₂, while 53% of the participants in the control group enhanced their achievement of LO₂.

Figure 8 delineates the relationship between the attainment of LO₁ and LO₂ within both groups. Undoubtedly, the contrast in outcomes between the experimental group and the control group provides compelling evidence that

firmly supports the pivotal role of the proposed framework in significantly improving SLOs.

Table 3 provides an analysis of survey results depicting the students' perspectives regarding the impact of the experiment on their motivation and dialectal variation. The findings suggest that a majority of participants in the experimental group concurred that the framework acted as a catalyst for enhancing their motivation to learn a foreign language, thereby fostering increased enthusiasm. Conversely, participants in the control group, who utilized the traditional learning method, exhibited lower levels of motivation, as evidenced by their performance in Tables 1 and 2. However, dialectal variation emerged as a challenge for participants in the experimental group.

In order to gauge the applicability and reliability of the framework, an online survey was administered to a cohort of eight English subject instructors, encompassing a single inquiry. Table 4 encapsulates a concise overview of the outcomes. Approximately 86% of the instructors expressed satisfaction with the framework, attributing their positive sentiment to its integration of learning outcomes and the incorporation of a point-based gamification aspect. This feature holds the potential to elevate students' motivation and engagement within the educational environment. However, 14% of instructors voiced reservations, advocating against the sole reliance on this framework for language instruction and proposing its integration with traditional teaching methodologies. From the student perspective, instructors highlighted concerns about certain students lacking access to devices, internet connectivity, or technical proficiency, potentially hindering their ability to engage with e-learning systems. From the instructor's viewpoint, it was emphasized that educators also require adequate training to navigate the technical intricacies of online learning. Moreover, dependency on such a framework for knowledge acquisition raises apprehensions about diminishing the primary role of teachers in imparting knowledge.

5. Discussion

This paper introduces a novel framework for learning foreign languages based on gamification and card games. In contrast to existing approaches that primarily emphasize elevating student motivation while neglecting the enhancement of SLOs, our primary objective was to improve the achievement of SLOs. Drawing on the findings of the case study presented in Section 4.2.6, the proposed solution demonstrated a favorable influence on the attainment of SLOs. Furthermore, it has been instrumental in augmenting student motivation to persevere through the educational journey, thereby offering a promising trajectory for the refinement of language instruction. However, the path to efficacious implementation necessitates a nuanced comprehension of the variations in dialectal representation. Moreover, instructors have expressed their endorsement of the use of our solution as a supportive adjunct to foreign language courses rather than exclusive reliance on it as the sole pedagogical tool for language acquisition. In addition to these findings,

TABLE 1: Cumulative achievement of the LO₁.

Knowledge and understanding (LO ₁)				Knowledge and understanding (LO ₁)			
Experimental group		Control group		Experimental group		Control group	
Student num	<i>t</i> ₁	<i>t</i> ₂	IR	Student num	<i>t</i> ₁	<i>t</i> ₂	IR
S1	70	73	4.29	S1	60	61	1.67
S2	81	84	3.70	S2	59	53	-10.17
S3	63	66	4.76	S3	70	70	0.00
S4	89	92	3.37	S4	62	64	3.23
S5	75	72	-4.00	S5	75	71	-5.33
S6	92	94	2.17	S6	66	64	-3.03
S7	77	82	6.49	S7	58	57	-1.72
S8	70	74	5.71	S8	83	85	2.41
S9	54	57	5.56	S9	75	70	-6.67
S10	62	65	4.84	S10	67	75	11.94
S11	81	80	-1.23	S11	77	80	3.90
S12	63	65	3.17	S12	70	70	0.00
S13	79	83	5.06	S13	54	56	3.70
S14	60	58	-3.33	S14	80	81	1.25
S15	71	68	-4.23	S15	63	61	-3.17
S16	64	66	3.13	S16	79	81	2.53
S17	77	75	-2.60	S17	74	70	-5.41
S18	80	80	0.00	S18	69	67	-2.90
S19	93	95	2.15	S19	62	60	-3.23
S20	72	74	2.78	S20	43	41	-4.65
S21	59	62	5.08	S21	83	85	2.41
S22	89	85	-4.49	S22	60	62	3.33
S23	95	98	3.16	S23	45	45	0.00
S24	87	85	-2.30	S24	88	86	-2.27
S25	92	90	-2.17	S25	68	70	2.94
S26	89	87	-2.25	S26	80	79	-1.25
S27	77	80	3.90	S27	75	77	2.67
S28	82	85	3.66	S28	78	80	2.56
S29	66	62	-6.06	S29	61	60	-1.64
S30	92	95	3.26	S30	56	58	3.57
S31	88	86	-2.27	S31	60	62	3.33
S32	90	93	3.33	S32	68	70	2.94
S33	84	82	-2.38	S33	60	62	3.33
S34	74	71	-4.05	S34	81	83	2.47
S35	63	65	3.17	S35	50	53	6.00
S36	75	78	4.00	S36	53	53	0.00
S37	84	89	5.95	S37	83	80	-3.61
S38	98	97	-1.02	S38	70	72	2.86
S39	52	55	5.77	S39	82	79	-3.66
S40	82	85	3.66	S40	68	70	2.94
S41	90	95	5.56	S41	55	53	-3.64
S42	65	68	4.62	S42	82	84	2.44
S43	82	85	3.66	S43	90	86	-4.44
S44	88	92	4.55	S44	71	69	-2.82
S45	82	80	-2.44	S45	89	87	-2.25
S46	75	78	4.00	S46	57	59	3.51
S47	54	58	7.41	S47	86	83	-3.49

TABLE 1: Continued.

Knowledge and understanding (LO ₁)				Knowledge and understanding (LO ₁)			
Experimental group		Control group		Experimental group		Control group	
Student num	<i>t</i> ₁	<i>t</i> ₂	IR	Student num	<i>t</i> ₁	<i>t</i> ₂	IR
S48	57	60	5.26	S48	79	81	2.53
S49	84	82	-2.38	S49	70	72	2.86
S50	92	94	2.17	S50	89	91	2.25
S51	93	97	4.30	S51	63	63	0.00
S52	90	93	3.33	S52	88	89	1.14
S53	62	64	3.23	S53	79	77	-2.53
S54	95	97	2.11	S54	83	85	2.41
S55	50	55	10.00	S55	88	96	9.09
S56	86	89	3.49	S56	74	70	-5.41
S57	80	83	3.75	S57	74	76	2.70
S58	62	64	3.23	S58	47	49	4.26
S59	78	80	2.56	S59	40	42	5.00
S60	77	80	3.90	S60	53	55	3.77

the proposed solution makes notable contributions when compared to prior research endeavors.

Fundamentally, the framework synergistically integrates the benefits derived from both gamification and card games to elevate the quality and effectiveness of the FLL environment. It integrates the point game mechanic and delivers the educational process within the context of a card game paradigm. The use of point game mechanics in e-learning systems contributes to increased motivation, provides a structured progress-tracking system, and provides a more enjoyable and engaging learning experience [46]. Moreover, the gamified nature of card games facilitates the acquisition of different language skills, enhances motivation, fosters active participation, and promotes a deeper understanding of linguistic concepts.

Besides, our framework surpasses prior approaches by offering a more comprehensive approach, with a specific emphasis on augmenting proficiency in the four fundamental language skills essential for foreign language mastery. As highlighted in Sections 2.4 and 2.5, a multitude of previous studies have utilized gamification or card games to cultivate proficiency in specific language components. For instance, the authors in [24] implemented game mechanics for time-limited w, while authors in [10] devised card game applications for the enhancement of English vocabulary skills in young learners.

Moreover, this study harnesses the capabilities of agent technology in the development of language learning systems, as the primary component of the proposed framework was the CDA. Integrating agents into e-learning systems creates a dynamic and responsive educational environment, ultimately elevating the quality and effectiveness of the learning process [47]. Besides, the framework employs the micro-learning method to deliver learning content. As illustrated in Section 2.3, microlearning assumes a pivotal role in the language learning process, serving as a crucial component

TABLE 2: Cumulative achievement of the LO₂.

Skills (LO ₂)				Skills (LO ₂)			
Experimental group		Control group		Experimental group		Control group	
Student num	<i>t</i> ₁	<i>t</i> ₂	IR	Student num	<i>t</i> ₁	<i>t</i> ₂	IR
S1	76	79	3.95	S1	66	66	0.00
S2	83	85	2.41	S2	60	60	0.00
S3	62	65	4.84	S3	69	72	4.35
S4	85	87	2.35	S4	63	66	4.76
S5	70	70	0.00	S5	70	68	-2.86
S6	90	95	5.56	S6	63	60	-4.76
S7	77	79	2.60	S7	56	54	-3.57
S8	72	76	5.56	S8	85	87	2.35
S9	60	62	3.33	S9	70	65	-7.14
S10	63	67	6.35	S10	62	59	-4.84
S11	80	77	-3.75	S11	72	70	-2.78
S12	65	67	3.08	S12	73	72	-1.37
S13	80	84	5.00	S13	56	52	-7.14
S14	63	67	6.35	S14	82	84	2.44
S15	77	80	3.90	S15	60	57	-5.00
S16	62	65	4.84	S16	80	82	2.50
S17	79	81	2.53	S17	72	69	-4.17
S18	80	84	5.00	S18	70	73	4.29
S19	95	98	3.16	S19	62	58	-6.45
S20	73	75	2.74	S20	47	49	4.26
S21	60	64	6.67	S21	81	79	-2.47
S22	88	90	2.27	S22	62	64	3.23
S23	93	95	2.15	S23	49	51	4.08
S24	84	87	3.57	S24	82	87	6.10
S25	90	95	5.56	S25	60	63	5.00
S26	90	91	1.11	S26	82	80	-2.44
S27	79	82	3.80	S27	72	75	4.17
S28	85	87	2.35	S28	72	70	-2.78
S29	65	68	4.62	S29	63	65	3.17
S30	95	97	2.11	S30	54	52	-3.70
S31	84	88	4.76	S31	58	60	3.45
S32	89	92	3.37	S32	67	70	4.48
S33	80	84	5.00	S33	63	65	3.17
S34	78	81	3.85	S34	80	78	-2.50
S35	67	69	2.99	S35	56	57	1.79
S36	79	78	-1.27	S36	55	57	3.64
S37	80	85	6.25	S37	85	87	2.35
S38	93	95	2.15	S38	74	76	2.70
S39	62	66	6.45	S39	80	83	3.75
S40	86	89	3.49	S40	63	65	3.17
S41	97	99	2.06	S41	57	55	-3.51
S42	68	66	-2.94	S42	79	76	-3.80
S43	84	86	2.38	S43	87	85	-2.30
S44	82	85	3.66	S44	74	72	-2.70
S45	80	80	0.00	S45	89	87	-2.25
S46	71	73	2.82	S46	55	56	1.82
S47	56	59	5.36	S47	88	89	1.14

TABLE 2: Continued.

Skills (LO ₂)				Skills (LO ₂)			
Experimental group		Control group		Experimental group		Control group	
Student num	<i>t</i> ₁	<i>t</i> ₂	IR	Student num	<i>t</i> ₁	<i>t</i> ₂	IR
S48	60	63	5.00	S48	78	76	-2.56
S49	88	90	2.27	S49	73	75	2.74
S50	93	97	4.30	S50	85	88	3.53
S51	92	95	3.26	S51	70	69	-1.43
S52	90	94	4.44	S52	88	89	1.14
S53	68	70	2.94	S53	75	72	-4.00
S54	96	98	2.08	S54	86	85	-1.16
S55	52	55	5.77	S55	88	91	3.41
S56	87	90	3.45	S56	71	68	-4.23
S57	79	75	-5.06	S57	74	76	2.70
S58	69	72	4.35	S58	49	51	4.08
S59	74	76	2.70	S59	46	48	4.35
S60	72	75	4.17	S60	59	61	3.39

in facilitating effective and targeted acquisition of linguistic skills [48].

In conclusion, this work represents a noteworthy advancement in the field of FLL. It introduces a novel approach to addressing the achievement of SLOs, which is deemed a pivotal aspect within the context of the language learning process. The presented approach to language learning integrates essential technologies, such as gamification, card games, agent technology, and the microlearning method. Through the synergistic integration of these components, our research contributes a multifaceted and pioneering solution to enhance the overall efficacy of GBL and FLL methods.

6. Conclusion

Gamification and card games are integral components of foreign language education, infusing the learning process with interactive and enjoyable elements. These educational tools not only serve to enhance student motivation but also offer practical applications for language skills, thereby fostering a dynamic and engaging language learning experience. However, current gamified or card game-based language learning approaches often prioritize motivating students while overlooking the achievement of SLOs. This gap is addressed in this paper, presenting a novel framework for learning a foreign language based on gamification and card games, aimed at enhancing the achievement of SLOs. The framework is developed based on agent technology and the microlearning strategy. The efficacy of the framework underwent evaluation through a case study. The primary focus was on assessing the impact of the proposed solution on students' achievement of learning outcomes. Additionally, a comprehensive evaluation of the framework was conducted, considering aspects such as motivation, dialect diversity, and applicability. The results revealed a positive influence of the proposed solution on augmenting students'

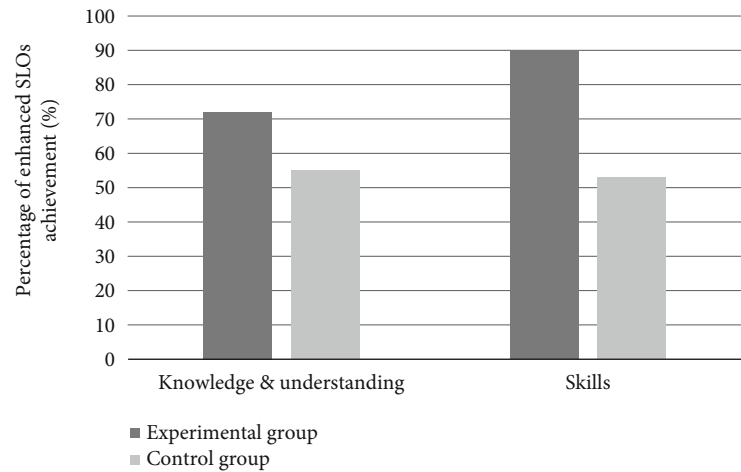


FIGURE 8: The achievement of SLOs in the experimental group versus the control group.

TABLE 3: Participants’ perspectives on motivation and dialectal variation.

(i) Would the instructional approach employed in the experiment enhance students' motivation to persist in their learning endeavors?						
Motivation		Strongly agree	Agree	Neutral	Disagree	Strongly disagree
	Experimental group	75%		15%	10%	0%
	Control group	45%		5%	35%	15%
(ii) Has the influence of your native dialect impacted language acquisition throughout the duration of the experiment?						
Dialect variation		Strongly agree	Agree	Neutral	Disagree	Strongly disagree
	Experimental group	55%		25%	10%	10%
	Control group	35%		45%	5%	15%

TABLE 4: Instructors’ perceptions on the applicability and reliability of the proposed framework.

Do you endorse adopting this framework to teach foreign language courses instead of traditional learning methods?				
Justify your answer.				
Strongly agree	Agree	Neutral	Disagree	Strongly disagree
85.6%		6.4%	8%	0%

achievement of learning outcomes and bolstering their motivation.

This work represents a significant contribution to the field of FLL, proposing an innovative approach for enhancing SLOs through the potential of gamification and card games. Additionally, it suggests leveraging the capabilities of AI and the microlearning method in implementing gamified or gameplay-based language learning environments. Despite these contributions, several limitations are identified. First, the study’s sample size is relatively small, primarily due to time constraints. The authors had to work with the participants who were available within a certain timeframe, and as a result, a larger pool of participants could not be accessed. Consequently, the study relies on a smaller sample size, potentially limiting the generalizability of the findings

to a broader population. Future research endeavors could consider employing strategies to address this limitation and enhance the study’s external validity, such as exploring collaboration opportunities with larger participant pools or extending data collection periods.

Secondly, dialectal variation among students posed a challenge. As indicated in Table 3, students speaking diverse dialects encountered difficulties in effectively engaging with the framework. Therefore, it is recommended to enhance the framework to accommodate a variety of native dialects. Thirdly, the study did not explore the framework’s applicability beyond traditional classroom settings. As a future endeavor, it is suggested to validate its effectiveness in supporting existing language learning platforms. Fourthly, the study employed only quantitative methodology. Incorporating qualitative methodologies in future research endeavors is recommended to complement and enrich the findings obtained in this study. Fifthly, the study investigated the influence of the framework on the attainment of learning outcomes specific to the English language discipline. The framework demonstrates a notable level of genericity, allowing for its application across diverse disciplines with minimal modifications. Its inherent flexibility and customizability enable users to tailor it to their specific needs and preferences. Additionally, the framework’s scalability

ensures its adaptability to various scales, complexities, and scopes, thereby enhancing its utility in diverse contexts. Consequently, future research will explore the framework's applicability to different academic disciplines to broaden its scope and efficacy.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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